Sequencing Design Considerations

Considerations before starting a project

? **What is the question you want to answer?**

? **What kind of data do you need to answer it?**

What is the question you want to answer?

- Is there a hypothesis you want to test?
- How many samples do you have?
    - Do you need more?
- Understanding this shapes many decisions

Number of reads/bases to sequence

- How much data do you need to generate to answer your question?
- This depends on the completeness and accuracy you want
- Decide that level before beginning the experiment — it determines how much data to generate
- Things you need to know
  - Genome size
  - Coverage level
  - Sequencing platform

An analogy

Trying a protocol that calls for 1 µg of DNA with only 0.1 µg may end up working — but it may not.

$$\text{\# Samples} = \frac{\text{Total Data from Sequencer}}{\text{Coverage Needed X Genome Size}}$$

Types of sequencing for Systematics/Phylogenetics

Genome assembly

- Reference genome
- RNA-seq (transcriptome)

Phylogenetic/Population genetic

WGS / WGRS

Whole-genome (re)sequencing

Reduced representation

- RAD-seq
- Ultraconserved elements (UCEs)
- Anchored hybrid enrichment (AHE)
- Amplicon sequencing

Reference genomes: check before you sequence

Step 0: search NCBI Datasets, ENA and GoaT (Genomes on a Tree) for your taxon.

Your species already has a chromosome-level genome?

Use it — after checking its quality (BUSCO, QV).

Only a close relative has one?

Mapping to it can work, but divergence causes reference bias and lost reads.

Nothing close?

Build your own — or ask whether a genome program already plans it.

3,400+

reference-quality eukaryote genomes by Sept 2024, from 500+ families (EBP)

150,000

species targeted in EBP Phase II, over ~4 years

~$6,100

projected cost per genome in Phase II (vs. ~$28,000 in Phase I)

Recipe for a reference genome in 2026

Sample

One individual, flash-frozen; high-molecular-weight DNA plus RNA

Long reads

PacBio HiFi (~25–30×) or ONT R10.4.1 long reads

Hi-C

Proximity-ligation short reads to link contigs into chromosomes

Assemble

Long reads → phased contigs (hifiasm; Verkko for combined data)

Scaffold & curate

Hi-C scaffolding (YaHS), then manual curation of the contact map

Annotate & submit

Gene models (BRAKER3, Helixer, NCBI EGAPx); deposit in NCBI/ENA

Rule of thumb: long reads build the contigs, Hi-C orders them into chromosomes, and RNA evidence finds the genes.

Long-read options for assembly in 2026

	PacBio HiFi	Oxford Nanopore
Read length	~15–20 kb	Standard ~10–30 kb N50; ultra-long >100 kb
Per-read accuracy	>99.9% (Q30+)	~99.75% simplex (Q26); duplex >Q30
Assembler	hifiasm	hifiasm (ONT) — new in 2026
Strengths	Mature pipelines; the Darwin Tree of Life / EBP standard	Longer reads; cheaper instruments; portable; methylation
Combined	HiFi + HiC +ONT ultra-long → telomere-to-telomere (Verkko, hifiasm UL)	

2026 UPDATE

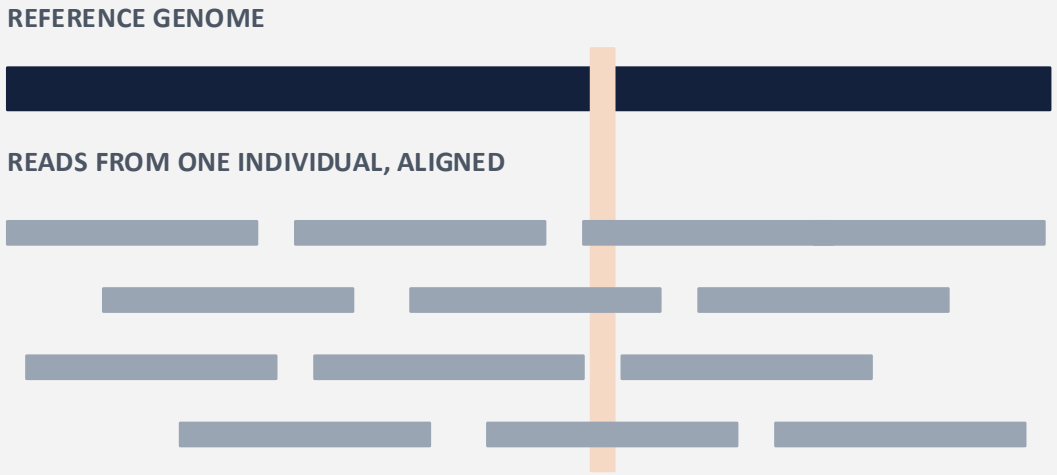
hifiasm (ONT) (Nature, Feb 2026) assembles near telomere-to-telomere genomes from standard R10.4.1 simplex reads — no ultra-long library needed, and it runs on ordinary CPUs.

Whole-genome resequencing: The Future is here

Assemble once, resequence many

1. Build (or borrow) one high-quality reference genome
2. Short-read sequence many individuals at modest coverage
3. Align each individual's reads to the reference
4. Differences from the reference become SNPs (and indels)

Result: millions of SNPs across coding and non-coding regions, with no probe or cut-site ascertainment.



Base differs from the reference in about half the reads covering this site → heterozygous SNP

Repeat for every individual, then genotype all of them jointly at every variable site.

The WGRS pipeline, from tissue to SNP matrix

Extract DNA

Tissue → genomic DNA; tens of ng to ~1 µg

Library & sequence

Barcoded libraries, pooled; 2 × 150 bp short reads

QC & trim

Remove adapters and low-quality bases (FastQC, fastp)

Map to reference

Align reads → BAM file (BWA-MEM2)

Clean alignments

Sort, mark PCR duplicates (samtools, Picard)

Call per sample

Genotype likelihoods → gVCF (GATK HaplotypeCaller; bcftools; DeepVariant)

Joint genotyping

All samples together → one multi-sample VCF (GATK GenotypeGVCFs)

Filter

Quality, depth, missing data, MAF → final SNP matrix

Tools shown are common examples, not the only choices. Steps 1–4 are shared with most short-read assays.

Calling a SNP: from pileup to genotype

Individual 1



0/0 homozygous reference

Individual 2



0/1 heterozygous

Individual 3



1/1 homozygous alternate

Top row = reference; "." = read matches the reference; G = alternate allele. Toy depth of 4 reads — real calls want ≥10×.

ONE ROW OF THE RESULTING VCF

```
CHROM   POS    REF ALT   QUAL   GT:DP
chr1    1008   A   G     812    0/0:4   0/1:4   1/1:4
```

Callers weigh base quality, mapping quality and depth into genotype likelihoods, then pick the most probable genotype.

How much coverage per individual?

Chance a heterozygote shows only one allele (and is miscalled homozygous):

50%
at 2× depth

6%
at 5× depth

0.2%
at 10× depth

0.00000000182%
at 30× depth

High coverage (≥10–30×)

- Confident individual genotypes, including heterozygotes and rare variants
- Fewer individuals for the same budget

Low coverage (1–5×)

- Many more individuals; analyze genotype likelihoods (ANGSD) or impute
- Good for allele frequencies, PCA, FST — not individual genotypes

Budget = genome size × coverage × individuals. e.g., 400 Mb butterfly genome × 10× coverage × 96 individuals ≈ 384 Gb ≈ 5% of one NovaSeq X 25B flow cell.

Filtering SNPs: what to keep

Site quality

Minimum QUAL / quality-by-depth (QD); drop low-confidence calls

Depth

Minimum and maximum — unusually deep sites often mark collapsed paralogs or repeats

Mapping

Mapping quality (MQ) and strand-bias (FS/SOR) filters

Missing data

Thresholds per site and per individual; drop poorly sequenced samples

Allele frequency

Biallelic sites only; MAF/MAC filters (keep singletons for demography)

Linkage

Thin or LD-prune for PCA, ADMIXTURE and methods assuming independent SNPs

Watch for: reference bias (reads from divergent taxa map poorly), paralogs collapsed onto one locus, and batch effects between sequencing runs.

From SNPs to Analyses

Question	Approach	Example tools
Relationships among individuals or populations	ML tree from SNPs with ascertainment-bias correction	IQ-TREE (+ASC), RAxML-NG
Species tree despite incomplete lineage sorting	Quartet-based coalescent methods	SVDquartets; ASTRAL on genomic windows
Population structure	PCA; admixture proportions	PLINK, ADMIXTURE
Introgression	ABBA-BABA (D-statistics), f-branch	Dsuite
Geneflow	Phylogenetic Networks	Treemix, Phylonet, Phylonetwork, Splittree
Demographic history	Coalescent inference from one diploid genome	PSMC, dadi, ABC
Divergence and selection	FST / dXY in sliding windows	VCFtools, pixy

Pulling phylogenetic loci from WGS data (Illumina)

Short-read whole genomes can stand in for capture data: extract the same loci in silico, then build gene trees and a species tree.

1 Assemble, then harvest

Assemble each sample de novo (SPAdes, MEGAHIT)

Search the assembly for target loci: BUSCO single-copy genes, or UCE / AHE probe sets

Best when: ~10–30× coverage; you want thousands of loci

Tools: BUSCO / compleasm; PHYLUCE (UCEs from genomes)

2 Assemble only the loci

Skip the full genome: use reference sequences as bait to pull and assemble matching reads

Much faster; works at lower coverage

Best when: many samples, a known locus set

Tools: HybPiper, aTRAM 2, Read2Tree

3 Map and call consensus

Map reads to a close relative's reference genome

Call variants; write a consensus sequence for each locus per sample

Best when: taxa are closely related to the reference

Tools: BWA-MEM2 + bcftools/samtools consensus

Low coverage (~1–5×, "genome skimming"): still enough for mitogenomes, nuclear rDNA and plastomes (GetOrganelle, MitoFinder).

Higher coverage (20-40X): Enough to pull Single copy orthologs

Watch for: paralogs, missing loci at low coverage, heterozygous sites in consensus sequences, and reference bias (route 3).

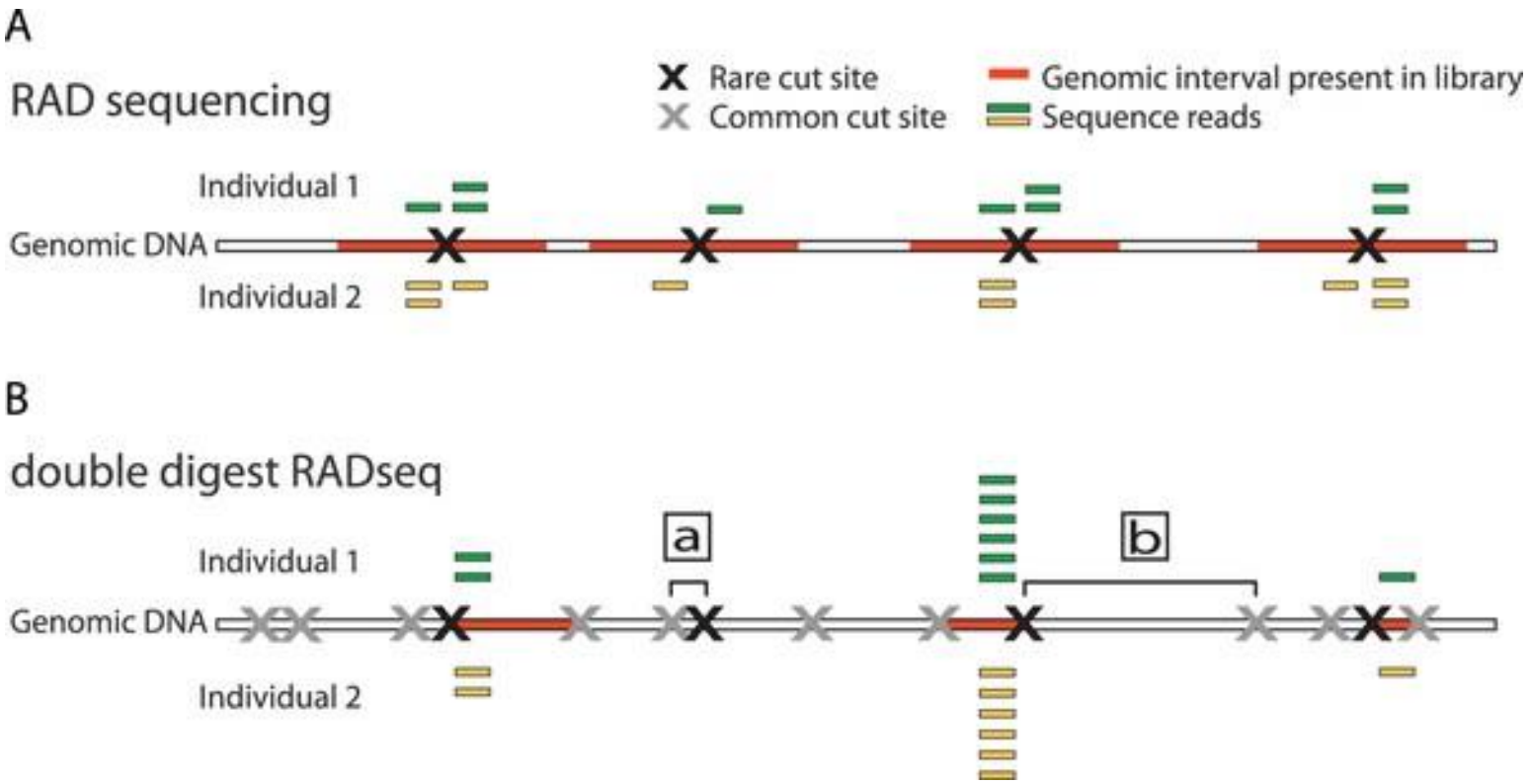
Harvested UCE/AHE loci can be combined with published capture datasets for the same probe set.

WGRS vs. reduced representation

	WGRS	RAD-seq	Target capture (UCE / AHE)
Loci	Whole genome; millions of SNPs, Thousands of Loci	Thousands of loci near cut sites	Hundreds to thousands of chosen loci
Reference genome needed?	Yes (same or close species)	No (de novo loci possible)	Probe set needed, not a genome
Cost per sample	Highest (scales with genome size) but getting cheaper	Low	Low–moderate
Library prep	Easiest	Moderate	Most complicated
Divergent taxa	Mapping gets harder with distance	Cut sites lost → missing data	Works across deep divergences
Degraded / museum DNA	Works at low coverage	Poor (needs intact fragments)	Good

Rule of thumb: WGRS for population-level questions within a species or genus with a reference; capture for deeper phylogenies and museum specimens.

RAD-seq



Restriction-site associated DNA

- Sequence only DNA next to restriction cut sites
- Same loci across many individuals
- ddRAD uses two enzymes plus size selection
- Baird et al. 2008; Peterson et al. 2012

RAD-seq library prep: original RAD vs. ddRAD

BAIRD ET AL. 2008

Original RAD

Digest with one rare-cutting enzyme (e.g., SbfI)

↓

Ligate P1 adapter carrying a sample barcode

↓

Pool samples; shear randomly

↓

Size select (~300–700 bp); ligate P2 adapter

↓

PCR (only P1–P2 fragments amplify) → sequence

PETERSON ET AL. 2012

Double-digest RAD (ddRAD)

Digest with a rare + a common cutter (e.g., EcoRI + MspI)

↓

Ligate barcoded adapters to both cut ends

↓

Pool samples

↓

Tight size selection picks a repeatable fragment window

↓

PCR with indexed primers → sequence

Shearing gives random P2 ends (lets you spot PCR duplicates). Double digestion gives fixed ends and lets you tune how many loci you get.

Analyzing RAD data: from reads to loci

Demultiplex

Sort reads by barcode; trim adapters and low-quality ends

Cluster within samples

Group similar reads into “stacks” (de novo) or map to a reference

Call alleles

Consensus alleles per locus per sample, above a minimum depth

Cluster across samples

Build a catalog of orthologous loci shared among individuals

Filter loci

Minimum samples per locus; drop loci with excess heterozygosity (paralogs)

Export

VCF, SNP matrix, or concatenated alignment for trees and structure

Tools: Stacks 2 (Rochette et al. 2019), ipyrad (Eaton & Overcast 2020). **Key settings:** clustering similarity threshold, minimum depth, minimum samples per locus.

RAD-seq pitfalls

Allele dropout

A mutation in the cut site removes that allele; heterozygosity is underestimated

Missing data grows with divergence

Cut sites are lost along branches, so missing data is phylogenetically structured (and can be informative)

PCR duplicates

Fixed-end ddRAD reads look identical; add degenerate bases or UMIs

Paralogs

Clustering can merge duplicated loci, producing false heterozygotes

Clustering threshold

Too strict splits alleles into separate loci; too loose merges paralogs

Depth of divergence

Best within genera or families; usable over tens of millions of years

See Gautier et al. 2013 (allele dropout); Rubin, Ree & Moreau 2012 and Eaton et al. 2017 (RAD phylogenetics and missing data).

Why target enrichment?

Choose loci in advance, then fish them out of each sample's library.

Same loci, every taxon

Orthologous loci recovered across deep divergences — families, orders and beyond

Degraded DNA works

Baits grab short fragments, so museum and herbarium specimens are usable

Cheap per sample

Pool 8–16 libraries per capture; 96 or more samples per run

Bonus data

Off-target reads often yield whole mitochondrial genomes

Two routes: PCR-based (amplicons; tens to hundreds of targets) and hybridization capture (hundreds to thousands of targets).

Target enrichment: PCR-based vs. hybrid capture

A

PCR-based enrichment

Genomic DNA

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Multiplex PCR with target-specific primers

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Amplified pool of regions of interest

↓

Add sequencing adapters to the amplicon pool

↓

NGS

B

Hybrid-capture enrichment

Fragment DNA; ligate adapters (library)

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Hybridize biotin-labeled probes

↓

Capture on streptavidin magnetic beads

↓

Wash off non-specific background

↓

Target-enriched DNA

↓

NGS

Hybridization capture, step by step

1. Build indexed libraries (shear to ~300–500 bp; old DNA is already short)
2. Pool libraries, often 8–16 per capture reaction
3. Add blockers (adapter and repeat DNA) and biotinylated baits
4. Hybridize baits to targets in solution
5. Pull down with streptavidin beads; wash away unbound DNA
6. Amplify the captured DNA and sequence

80–120 nt
biotinylated RNA or DNA baits

~16–24 h
hybridization; lower temperature for divergent taxa

~20–70%
of reads on target (varies widely)

Designing a probe set

Mine genomes or transcriptomes from a few taxa across the group

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Find conserved, single-copy loci present in all of them

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Design baits: tile ~120 nt across each target; add divergent variants

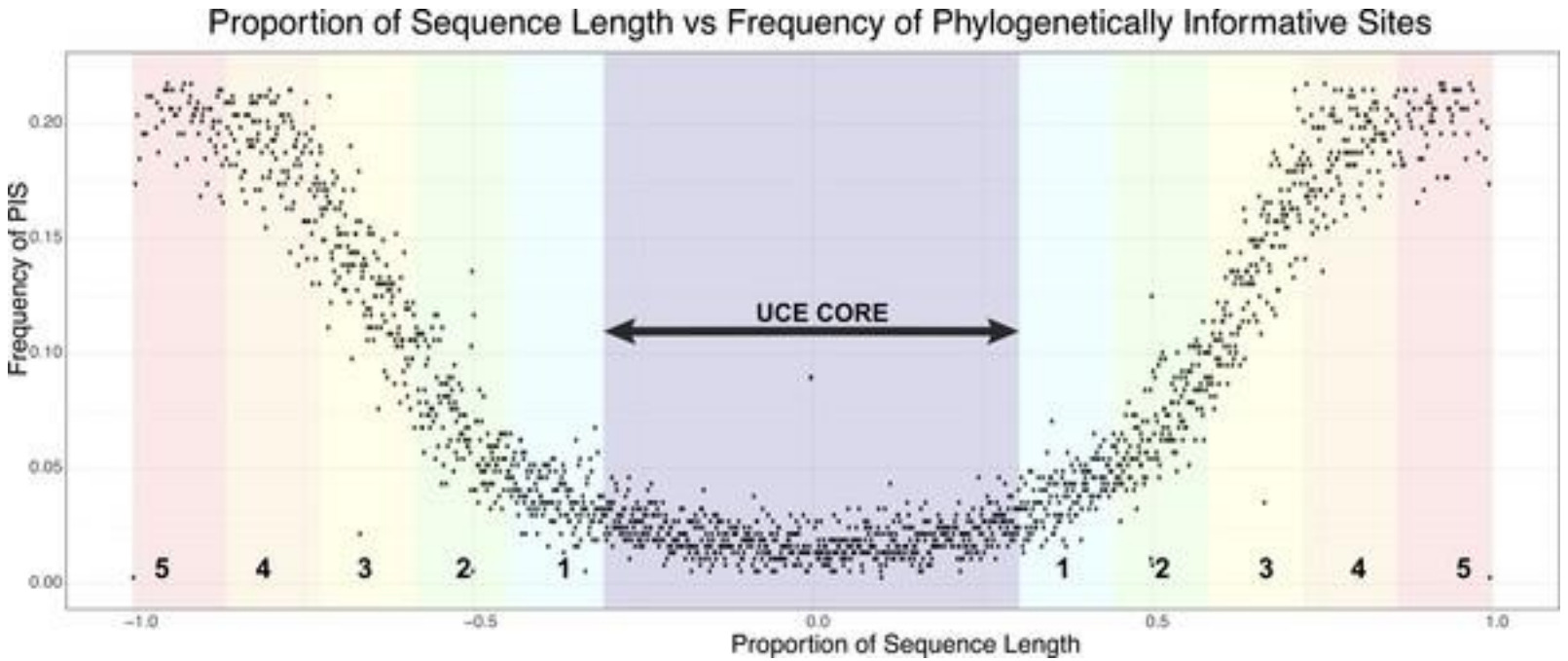
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Test in silico, then on a pilot plate

Probe set	Loci	Group
UCE Tetrapods 5K	~5,000	Tetrapods
UCE Hymenoptera v2	~2,600	Ants, bees, wasps
AHE vertebrate (Lemmon et al. 2012)	~500	Vertebrates
LEP1 AHE (Breinholt et al. 2018)	855	Lepidoptera
Angiosperms353 (Johnson et al. 2019)	353	Flowering plants

Use an existing kit when one fits your group — results then combine across studies.

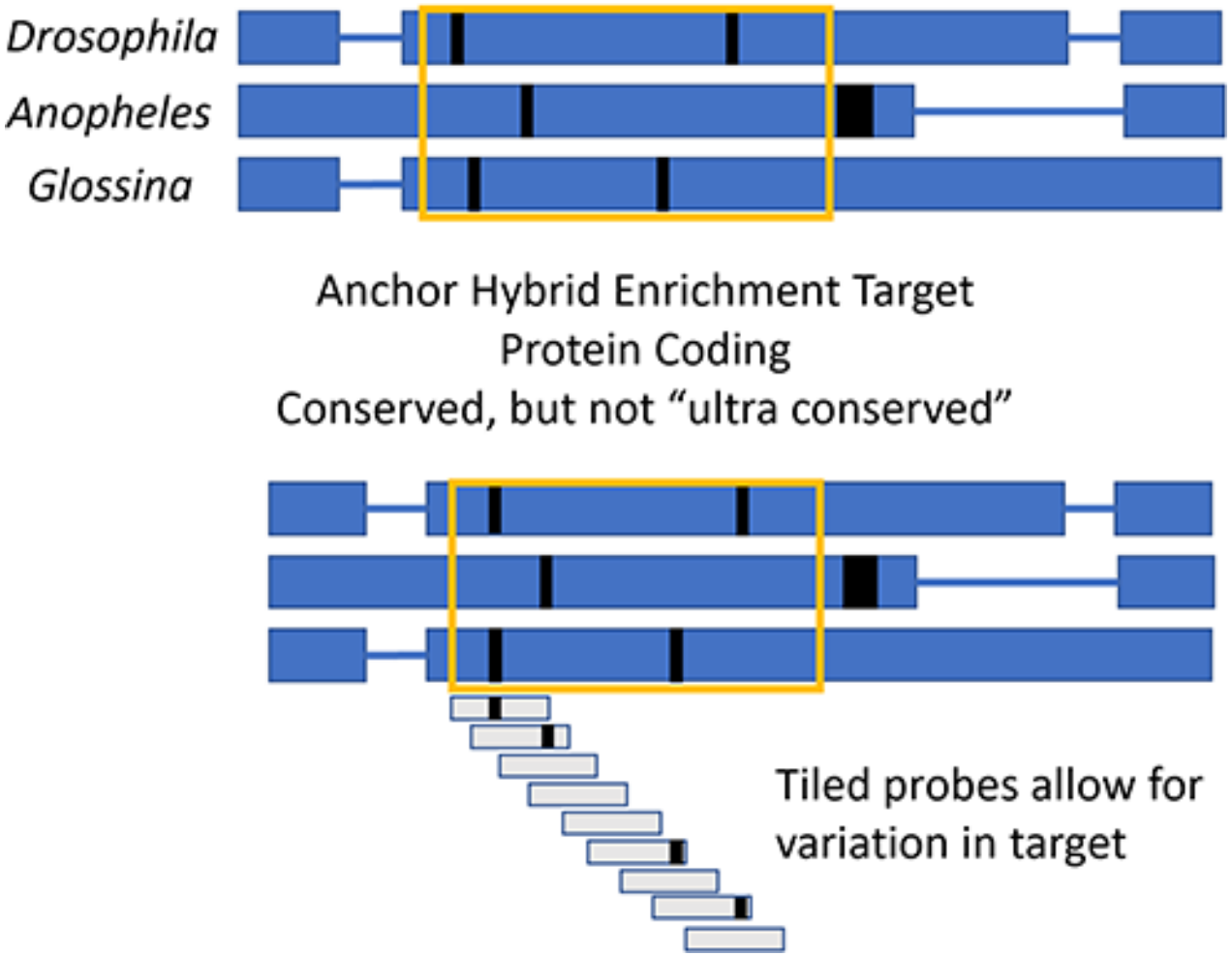
Ultraconserved elements (UCEs)



Signal lives in the flanks

- Highly conserved core anchors the probes
- Variability rises with distance from the core
- Faircloth et al. 2012

Anchored hybrid enrichment



Conserved, not “ultra-conserved”

- Targets protein-coding anchor regions shared across a clade
- Tiled probes tolerate variation within the target
- Flanking regions add variable sites
- Lemmon et al. 2012

Processing capture data

Demultiplex & trim

Sort by index; remove adapters (short inserts read into them)

Assemble per sample

Reads → contigs for each sample (SPAdes, Trinity)

Match to targets

Identify contigs for each locus; flag possible paralogs

Extract loci

Pull target regions plus variable flanking sequence

Align & trim

Multiple sequence alignment per locus (MAFFT); trim poorly aligned ends

Infer trees

Gene trees (IQ-TREE) → species tree (ASTRAL), or concatenation

Pipelines: PHYLUCE for UCEs (Faircloth 2016), HybPiper (Johnson et al. 2016), SECAPR (Andermann et al. 2018).

Capture pitfalls, and why museums love it

Bait divergence

Capture efficiency drops as targets diverge from the baits

Paralogs

Multi-copy genes can masquerade as orthologs; use paralog warnings

Cross-contamination

A risk with low-input and old samples; include extraction blanks

DNA damage

Old DNA shows C→T changes at read ends; trim or model them

MUSEOMICS

Century-old specimens are sequenceable

Because baits capture short, fragmented DNA, pinned insects, skins and herbarium sheets — including type specimens — can be placed on the tree alongside fresh samples.

Choosing a platform in 2026

Goal	Typical approach	Platform
Many samples, SNPs or loci	RAD-seq, UCE/AHE capture, low-coverage WGS	Illumina, Element, Ultima, MGI
Degraded or museum DNA	Capture or shotgun on short fragments	Short-read platforms
Reference genome	HiFi + Hi-C (± ultra-long nanopore)	PacBio + Illumina Hi-C (+ ONT)
Structural variants, phasing, methylation	Long-read WGS	PacBio HiFi or ONT
Field work or fastest turnaround	Portable or same-day sequencing	ONT MinION; Roche SBX (new)